STAT818G Final Report

Review of *Early-predicting dropout of university students: an application of innovative multilevel machine learning and statistical techniques*

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1 Introduction

1.1 Problem Formulation

Education is a complicated and constantly evolving field with vast data generated from various sources. The rapid expansion of online education and the broadening of institutional management systems, and thus the growing complexity and size of data, have been key factors in a shift toward cutting-edge statistical techniques. The data often has irregularities, nonlinearities, and high dimensionality, causing problems for traditional methods. Robust machine learning (ML) approaches exploit the natural dynamic and multidimensional patterns of real-world educational systems, making them extremely flexible in solving a wide range of educational problems with the potential to improve the accuracy of results. Some pertinent examples are identifying and characterizing factors that influence student performance, optimizing adaptive learning technologies, and predicting at-risk students. A particular use case is the early prediction of dropout for university students, a topic explored by Cannistra et al (2022), which is reviewed here. Their paper uses multilevel modeling in conjunction with random forests in an effort to exploit the inherent structure of education data while maintaining robustness to complex, nonlinear relationships. Importantly, ML results are compared with generalized linear models. In the end, several critiques and avenues for future work are provided.

1.2 Motivations

There are clear motivations for predicting student dropout, summarized in terms of equity and efficiency. For the former, the authors note that academic literature has confirmed that disadvantaged students (from a socioeconomic perspective) are more at-risk of dropping out. Efforts to address this inequity have had limited success as of the publication of their paper. For the latter, student dropout represents a waste of resources because educating students is an expensive task, and the benefits of increased human capital via credentials acquired through education are not realized. Incorrectly predicting an at-risk student is an additional inefficiency, which may lead to the misallocation of resources and exclusion of genuinely struggling students. These issues place significant emphasis on the importance of generating empirical results that can be used for effective, data-driven decision-making. If at-risk students are identified with sufficient precision, individualized tutoring and intervention systems can be implemented.

2 Methods

2.1 Definitions and Data

The authors define dropout as the occurrence of any student who left the institute for reasons other than graduation. They also specified between early and late dropouts, because each may have different driving factors. The former occurs when a student drops out within the third semester after enrollment, while the latter occurs any time afterward.

The data were administrative data from Politecnico di Milano (PoliMi), which included various cohorts of first-year students over 9 years. It was initially comprised of over 10 million entries, each a representation of an administrative event or a student's set of features. After selecting only ended careers at the institution and reformatting to have students as the unit of analysis, the final sample included 31,071 students from 2010 to 2016 over 20 degree courses. Of these students, 62.7% graduated, 21.7% dropped out early, and 15.6% dropped out late. Based on previous literature, the selected covariates were the student's gender, income range, age, origin, high school track, admissions test grade, total credits earned in first semester, and number of exam attempts in the first semester. The algorithms discussed below are contextualized with these variables. Those with the strongest predictive power are of particular analytic and stakeholder interest.

2.2 Models

The outcome variable is binary, equal to 1 when the student dropped out and 0 otherwise. It can therefore be considered as a Bernoulli random variable with its parameter representing the probability of dropping out. This motivates the use of three techniques built on logistic regression, which are outlined below. For each model, 70% random subsets of the data were used for training, with the remaining 30% used for testing. The authors implemented each technique in three different ways: a) without degree courses, b) with dummy variables representing the degree course of the given student, and c) with degree courses as a random effect (multilevel modeling), resulting in 9 total models.

2.2.1 Generalized Linear Multilevel Model (GLMM)

This approach captures the variability in dropout due to students taking different degree courses, assuming dependence between students in the same course and discrepancies between students in different courses. GLMM allows researchers to distinguish between student- and course-level variability, exploit the hierarchical data structure, and incorporate the student-level covariates as fixed effects. The two-level model is

$$
\begin{aligned}
p_{ij} &= \mathbb{E}[Y_{ij}|b_i]; \quad j = 1, ..., n_i; \quad i = 1, ..., N \\
logit(p_{ij}) &= \sum_{k=1}^{K+1} \beta_k x_{ijk} + \sum_{q=1}^{Q+1} b_{iq} z_{ijq}; \quad b_i \sim N(0, \Psi)
\end{aligned}
\tag{1}
$$

where Y_{ij} represents the dropout outcome of the jth student in the ith course. β_k contains the coefficients for the K student-level covariates x_{ij} along with the overall intercept (fixed effects). The vector b_i represents the random effect for the ith course, its $Q+1$ predictors contained in z_{ij}. Ψ is a $(Q+1) \times (Q+1)$ within-group covariance matrix of the random effects coefficients. Note that this model assumes all Y_{ij} to be independent.

2.2.2 Generalized Mixed-effects Regression Trees (GMET)

GMET extends the GLMM by replacing the fixed effects part with a tree structure, placing it in the ML setting. This approach lacks any assumption of a particular functional form, making it extremely flexible and better for modeling interactions among covariates. The tree structure can be denoted as $f(x_{ij})$, a partition of the covariate space into rectangles where the prediction within each box is the mode of all observations within the rectangle. The regression tree algorithm defines the boxes by minimizing the variability within them and maximizing the variability between them. Covariate selection is achieved by virtue of the algorithm splitting the data in the most relevant manner.[1] The resulting structure gives insights into interactions among covariates as well.

2.2.3 Generalized Mixed-effects Random Forest (GMERF)

GMERF replaces the fixed effect part of GLMM with a random forest. Random forests extend the ideas of regression trees by taking many training sets from the entire population and building a separate prediction model using each training set. The resulting predictions are then averaged together, which reduces model variance and multicollinearity issues.[2] Other advantages are the generation of feature importance scores based on the corresponding mean decrease in Gini index, along with marginal associations between each predictor and the outcome.

3 Results

Model performance was compared by area under the ROC curve, with additional emphasis on accuracy, sensitivity (critically, the correct identification of at-risk students), and specificity. The 9 models were fit to early and late dropout students separately, with results summarized in Tables A.1 and A.2, respectively. The multilevel models (c) performed better than the non-nested models (a) in terms of AUC and sensitivity. All models were markedly better at identifying early dropout than late dropout, evidenced by the decrease in each metric. In terms of AUC and accuracy for early dropout, multilevel random forest performed slightly better than the classification tree, but failed to outperform GLM. However, both ML methods had greater sensitivity than GLM under multilevel specifications. For late dropout, the greatest sensitivity was attained by the dummy variable specification for classification trees, which appeared to be an outlying value considering the others. In general, the results provide strong evidence in favor of degree courses as a random effect, notwithstanding its improvement to interpretability. More evidence is provided by the notable variance partition coefficient (VPC) values, representing the portion of variability in the response explained by the course-level grouping. However, the differences between the classical and ML multilevel methods appear small.

In terms of covariates, the primary takeaways are as follows: i) males are more likely to late dropout than females, ii) native Italians off-site are more likely to early drop than Italians in-site, iii) students starting at older ages are more likely to late drop, iv) students who attended "Other" and technical high schools are more likely to late drop than those who attended academic, scientific high schools, v) students with higher admissions test scores are more likely to dropout early, vi) earning more first semester credits lowers

[1]That is, with the greatest discriminatory power (lowest Gini index).

[2]Representing a reduction of overfitting.

both early and late dropout probabilities, vii) multiple exam attempts in the first semester decreases likelihood of early drop but increases likelihood of late drop, relative to students with one attempt, viii) students without any exam attempts during the first semester are more likely both to early and late drop, ix) less advantaged students (in terms of income group) are less likely to early drop than their more advantaged counterparts.

The variable importance plots obtained by the multilevel random forest models are shown in Figure A.1. The number of credits obtained in the first semester is the most important variable by a considerable margin for both early and late dropout prediction. The authors demonstrate that it is the *only* covariate used to build trees. That is, the algorithm does not split on any of the other student features when predicting dropout. The authors emphasize that despite this finding, other covariates should not be considered irrelevant. They found that numerous other factors, such as nationality, residence, admission score, and income are significant predictors of the number of first semester credits. There may be a structural dependence between these factors worth exploring.

The first semester credits variable is largely responsible for the minimal performance differences between ML and classical methods. Their partial dependence plot (PDP, Figure A.2) shows that there is a largely linear relationship between the outcome and the number of credits earned in the first semester. Essentially, the predictor has a strong linear relationship with student dropout, so its inclusion in the models nullifies the strengths of the ML approaches in compensating for *nonlinearities*.

4 Discussion

The paper serves as an excellent starting point for analyzing student dropout with machine learning. Multilevel models were established as an appropriate and effective method in this context, which can be extended to other educational settings. However, ML methods did not significantly outperform classical methods in this example, largely due to the strong linear relationship between a specific predictor and the outcome. This section discusses some theory behind this result, argues for the continued use of ML in the educational context, and lists potential avenues for future research.

A primary takeaway is that institutions should invest heavily in *early* student success and productivity, as a successful first semester in terms of credits earned is by far the most important predictor of dropout. The linearity between first semester credits and dropout probability revealed by their PDP explains why ML performed similarly to traditional methods, since the nonlinear capabilities of ML are unnecessary in the presence of the credits feature. PDPs are in general an effective way of assessing ML validity and can be used to advise stakeholders on which models should be utilized to boost efficiency and resource allocation, but they can be limited. They assume independence between features, which may not be applicable in this case considering the statistical significance of nationality, income, admissions scores, etc. in predicting first semester credits. They may also be limited in that average marginal effects can obfuscate relationships that vary for different sets of data (that is, different types of students). More analysis is needed in their work to confirm the implications of the PDPs, including the distributions of features beneath the plot. The PDPs should also be stratified by various subsets of students (e.g. socioeconomic or demographic factors) in order to determine which factors are contributing most to the linear relationship. This would help researchers determine which contexts and settings make ML favorable to GLM.

Although this particular research did not demonstrate its strengths over traditional methods, ML should still be a part of educational research for the reasons discussed in the introduction. The initial motivations behind using ML are critical, as many problem settings do not satisfy linearity or independence assumptions or require robustness to increasingly complex data. From a policy perspective, even if ML does not provide a large improvement over classical methods, predicting just *one* more at-risk student accurately is worth the investment. Sensitivity is the most important aspect in this case, which was *maximized by tree methods* for both types of dropout. Further, the multilevel random forest approach can certainly be applied earlier, when the number of first semester credits is not available. ML offers much more flexibility in terms of timing. Since early intervention is critical to effectively assist students, this is a clear motivation for future research. Other ideas are summarized into two categories below.

- Potential improvements
  - Hyperparameter tuning should be reported, namely on the number of trees, tree depth, and number of features for random forests. The extra trees algorithm may be tested to further reduce overfitting and ensure computational feasibility, especially if the methods are to be scaled to more complex data.
  - An assessment of data imbalance is needed, since certain student groups may be over- or under-represented which can lead to biased conclusions. Propensity score analysis (or stratified sampling for model training sets) can ensure reliability and strengthen applications to equity concerns while addressing the potential shortcomings of PDPs.
  - Sensitivity analysis would be useful, especially as it relates to the inclusion of the number credits in first semester. Since this predictor dominated the models, what happens to model performance if it is the *only* predictor? Conversely, what happens to model performance if it is removed? What are the real-world consequences of either of these approaches?
- Extensions
  - Including data with courses in other subjects at different institutions. A three-level hierarchical approach with students nested within courses nested within institutions may be effective. Time-varying covariates and engineered features may also be of interest.
  - Student performance is a complicated outcome that may not be accurately reflected by a single value. A categorical or continuous outcome, or perhaps multiple outcome metrics simultaneously, may be worth considering.
  - The authors note that some of their results conflict with other literature, which may motivate the use of meta-analyses to reconcile discrepancies.
  - Any extensions would benefit from a detailed reporting of computation times and scalability to ensure real-world value and improve stakeholder interest.

The most important aspect of ML in the educational context is providing *actionable insights* to stakeholders, such as educators, administrators, and policy makers. Future work in this field must be guided by similar motivations as quality and access to education improve around the world. The ultimate goal is to determine the optimal balance between model performance, speed, and feasibility so that students are successful.

References

[1] Cannistrà, Marta et al. "Early-Predicting Dropout of University Students: An Application of Innovative Multilevel Machine Learning and Statistical Techniques." Studies in higher education (Dorchester-on-Thames) 47.9 (2022): 1935–1956. Web.

[2] Molar, Cristoph. "Interpretable Machine Learning: A Guide for Making Black Box Models Explainable". Chapter 19. https://christophm.github.io/interpretable-ml-book/pdp.html. Web.

[3] Swapnil B. Mohod. "Applied Nonlinear Analysis of Machine Learning Models for Education Streams: A Mathematical and Statistical Perspective." Communications on Applied Nonlinear Analysis 32.8s (2025): 75–86. Web.

Appendix

Table A.1: Fit statistics for all models in predicting early dropout

Model	Not nested (a)	Dummy (b)	Nested (c)
Generalized Linear Model (1)			
AUC	0.9576	0.9614	0.9615
Accuracy	0.9178	0.9219	0.9224
Sensitivity	0.8943	0.8913	0.8921
Specificity	0.9234	0.9291	0.9296
VPC			0.1063
Classification Tree (2)			
AUC	0.8748	0.8748	0.9473
Accuracy	0.9342	0.9342	0.9118
Sensitivity	0.7789	0.7789	0.9004
Specificity	0.9708	0.9708	0.9145
VPC			0.0857
Random Forest (3)			
AUC	0.9512	0.9553	0.9598
Accuracy	0.9183	0.9155	0.9160
Sensitivity	0.8709	0.8898	0.8966
Specificity	0.9294	0.9216	0.9205
VPC			0.1803

Table A.2: Fit statistics for all models in predicting late dropout

Model	Not nested (a)	Dummy (b)	Nested (c)
Generalized Linear Model (1)			
AUC	0.8977	0.9089	0.9091
Accuracy	0.8637	0.8593	0.8590
Sensitivity	0.7634	0.8003	0.7979
Specificity	0.8855	0.8721	0.8723
VPC			0.0852
Classification Tree (2)			
AUC	0.8714	0.8902	0.9049
Accuracy	0.8851	0.8157	0.8393
Sensitivity	0.6730	0.8718	0.8118
Specificity	0.9312	0.8035	0.8453
VPC			0.1193
Random Forest (3)			
AUC	0.8897	0.9016	0.9065
Accuracy	0.8640	0.8519	0.8549
Sensitivity	0.7354	0.7880	0.8036
Specificity	0.8919	0.8658	0.8660
VPC			0.1276

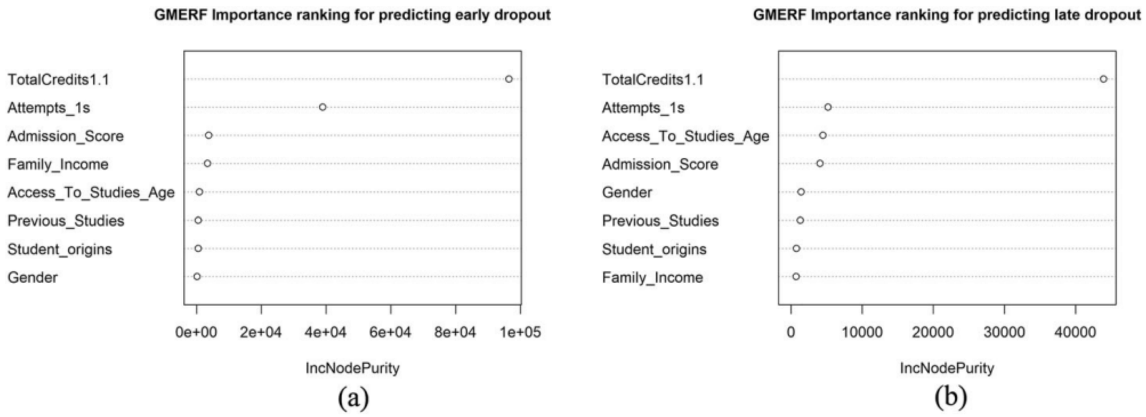

Figure A.1: Variable importance plots via random forest for (a) early and (b) late dropout prediction

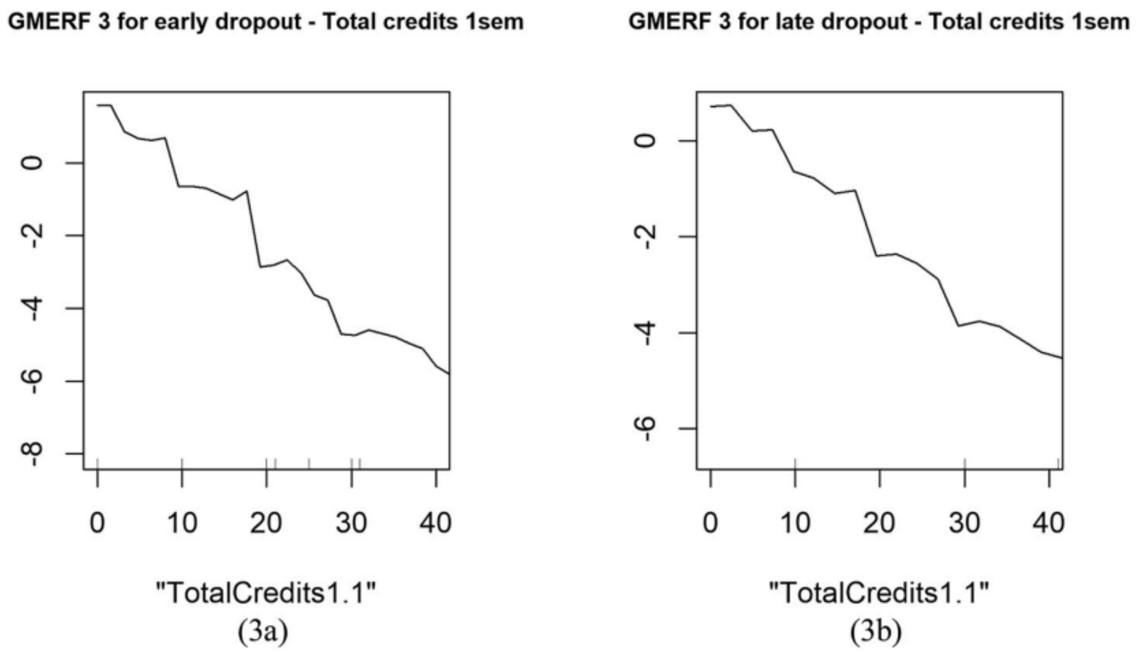

Figure A.2: Partial dependence plot for multilevel random forest model - number of credits earned in first semester. (3a) early and (3b) late dropout prediction